CHITRALEKHA BAIRAGI

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in full-stack backend development and Generative AI. Proficient in C++, Python, Data Structures & Algorithms, and REST API design with hands-on experience integrating LLMs, databases (MySQL, MongoDB), and cloud environments (AWS, Docker). Strong background in designing modular, scalable systems following Agile development practices.

TECHNICAL SKILLS

- **Core Languages:** Python, C++, Data Structures & Algorithms (DSA), OOP, DBMS
- **Backend & Web Development**: FastAPI, Flask, REST APIs, Streamlit
- **AI / Machine Learning:** Machine Learning, Deep Learning, Scikit-learn, TensorFlow, Model Evaluation
- **Generative AI & NLP:** Generative AI, Large Language Models (LLMs), RAG, Prompt Engineering, LangChain, Langgraph , MCP, Hugging Face, Sentence Transformers
- **Databases & Vector Stores:** MySQL, MongoDB, ChromaDB
- **Cloud & DevOps:** AWS, Docker, Git, GitHub

EXPERIENCE

Medicaps Univeristy | AI-Powered Multimodal Hospitality Creator (RAG + Image Generation)

Jan 2026 – April 2026

Python, Flask, ChromaDB, REST APIs, OOP | GitHub

- Designed and implemented an end-to-end backend pipeline with a RESTful API layer, integrating ChromaDB and Sentence Transformers for embedding-based semantic retrieval to ground generated content and reduce hallucination.
- Built object-oriented, modular Python services connecting an LLM (Gemini) with Stable Diffusion for multimodal content generation, structuring the codebase into independently testable retrieval, generation, and image modules.
- Deployed the pipeline via a production-ready Gradio UI and version-controlled the project on Git, iterating through multiple debugging cycles to stabilize the RAG-to-image generation flow end to end.

Medicaps Univeristy | Multi-Agent Trip Planner

Jan 2026 – April 2026

Python, CrewAI, LangChain, Groq API | GitHub

- Built a multi-agent system with CrewAI, coordinating 3 specialized agents (research, itinerary planning, and budgeting) via clean, well-defined API contracts to jointly produce a coherent travel plan.
- Integrated real-time web search (DuckDuckGo) with the Groq API for low-latency LLM inference, keeping agent responses fast and grounded in current information rather than stale training data.

PROJECTS

TrafficEye AI – Traffic Violation Detection & Analytics

Python, YOLOv8, OCR, Streamlit, SQLite, OOP | GitHub

- Architected a 4+ module end-to-end software pipeline — detection, OCR, evidence generation, and analytics — using object-oriented design, automating the full workflow from live video input to structured database storage.
- Integrated a YOLOv8 detection model with an OCR engine to extract license-plate data and auto-generate timestamped evidence records for each flagged violation.
- Built a Streamlit + SQLite dashboard for real-time monitoring and analytics, iterating through multiple debugging and QA cycles to keep each module independently testable and swappable.

AI Code Reviewer Agent

Python, LLMs, Webhooks, REST APIs | GitHub

- Built an autonomous, webhook-driven code review agent that listens for repository events and triggers automated LLM-based review via REST API-driven automation.
- Designed an interactive dashboard to surface review findings and history, cutting manual review turnaround time by an estimated 30–40% in testing.

EDUCATION

Medicaps University — Indore, MP

Aug 2022 – May 2026

B.Tech in Computer Science Engineering | CGPA: 8.7/10

CERTIFICATIONS

- Agentic AI & Generative AI – NASSCOM / Datagami (IBM Gold Partner), 80-hr Skill-Based Projects (Jan – Mar 2026)
- AWS Academy Graduate – Cloud Foundations & Cloud Architecting (Jun – Jul 2025)
- Exploratory Data Analysis with Python & GenAI (November 2025)

ACHIEVEMENTS

- Top 50 out of 300+ participants, University-Level Round, Smart India Hackathon (SIH) 2025.
- Built and shipped agentic AI applications for the "Agents League – Creative Apps" challenge.